



**UTM**  
UNIVERSITI TEKNOLOGI MALAYSIA

SECJ2203: Software Engineering

## **System Documentation (SD)**

Project Title

Version 1.0

Date: 1/12/2024

Faculty of Computing

Prepared by: AlphaTech

## Revision Page

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### a. Overview

The current version of the System Documentation (SD) covers on the Software Requirement Specification which is covered until Chapter 2.

### b. Target Audience

This document is intended for specific readers like KADA's cooperative, staff, and administrative. The rest of this SRS will contain the description of the product, external interface requirement, system features, nonfunctional requirement and other necessary requirements. The reader can start with the second main topic which is overall description and proceed through the end. Meanwhile, developers can go through external interface requirements for time saving.

### c. Project Team Members

| Member Name | Role        | Task | Status     |
|-------------|-------------|------|------------|
| Iman Abadi  | Team Leader |      | incomplete |
| Alif Fathi  | Team Member |      | incomplete |
| Afif Shaqir | Team Member |      | incomplete |
| Afiq Danial | Team Member |      | incomplete |

### d. Version Control History

| Version | Primary Author(s)                          | Description of Version | Date Completed |
|---------|--------------------------------------------|------------------------|----------------|
| 1.0     | Team leader 1 (IMAN ABADI BIN MOHD NIZWAN) | Completed Chapter 1& 2 | 28/11/2024     |
|         |                                            |                        |                |
|         |                                            |                        |                |

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# 1. Introduction

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## 1.1 Purpose

The purpose of this document is to present details of the Web-based Platform for Kemubu Agricultural Development Board or KADA's system. It will explain the purpose and features of the system, the interfaces of the system, what the system will do, the constraints under which it must operate and how the system will react to external stimuli. This document is also intended for both KADA's cooperative staff and our developers of the system. This will be proposed and presented to the KADA's cooperative representative.

## 1.2 Scope

The scope of this project is to propose a system that will be a website that replaces manual systems by using multiple technologies to ensure the smooth-running system for the cooperative. Each of this technology has its own functionality to help the development of this system. The system will have some existing functionality but with additional changes using these technologies will increase efficiency to enhance the working experience because the technologies used will develop a web server with a relational database management system. The system will cover functionalities such as user authentication, membership application, membership application, administrative processes and successful payment notification. The system will not cover scopes that utilize payment gateways for tasks such as making a payment through the system, buying share modal and claiming of member benefits.

### 1.3 Definitions, Acronyms and Abbreviation

- KADA: Kemubu Agricultural Development Board
- HTTPS: Hypertext Transfer Protocol Secure
- SRS: Software Requirement Specification
- HTML: Hypertext Markup Language
- iOS: iPhone Operating System
- SSL: Secure Sockets Layer
- TLS: Transport Layer Security
- SMTP: Simple Mail Transfer Protocol
- RAM: Random Access Memory
- AES: Advanced Encryption Standard

### 1.4 References

TBD

### 1.5 Overview

Chapter 2 will be covering the specific requirements of the system. The user of the system is identified including the task they are involved with. The tasks are then detailed so that it can be better understood.

## Specific Requirements

Chapter 2 will be covering the specific requirements of the system. Personas are created to identify the task the users of the system needs to perform. The system features were covered after by using a use case diagram to illustrate the tasks present in the system. The use cases are further detailed by using use case descriptions, sequence diagrams and activity diagrams to avoid ambiguity of the task involved.

### 1.6 Persona

Personas are intended users of the system and their characteristics, such as their technical expertise, knowledge, and physical abilities. This section provides the description of the user of the system determining their needs and making a story out of the user needs to break down the requirements of the system.

#### 2.1.1 Persona 1: KADA's Employee

##### User Need

- KADA's employees need a way to become a member of the KADA to be eligible to apply for a loan from the organization.

##### User Stories

- As a KADA's employee, I want to apply for a personal loan through the KADA for my personal financial needs.
- As a KADA's employee, I have to register as a member of KADA with my own KADA's official email.

#### 2.1.2 Persona 2: KADA's Member

##### User Need

- KADA members need a way to apply for a loan and access the dashboard.

##### User Stories

- As a KADA member, I want to apply for a loan so I can borrow a large amount of money.
- As a KADA member, I want to log in the system and check my debts in the dashboard.

### **2.1.3 Persona 3: System's Administrator**

#### **User Need**

Administrators need a way to manage data contained within the system's database and provide financial statements for both members and employees.

#### **User Stories**

- As a system's administrator, I want to be able to log into the system so that I can manage the data in the system's database.
- As a system's administrator, I want to be able to provide financial statements for both members and employees so that they can view their financial reports.

### **2.1.4 Persona 4 : KADA's Board of Director**

#### **User Need**

Board of Director needs a way to approve the loan application and access the dashboard.

#### **User Stories**

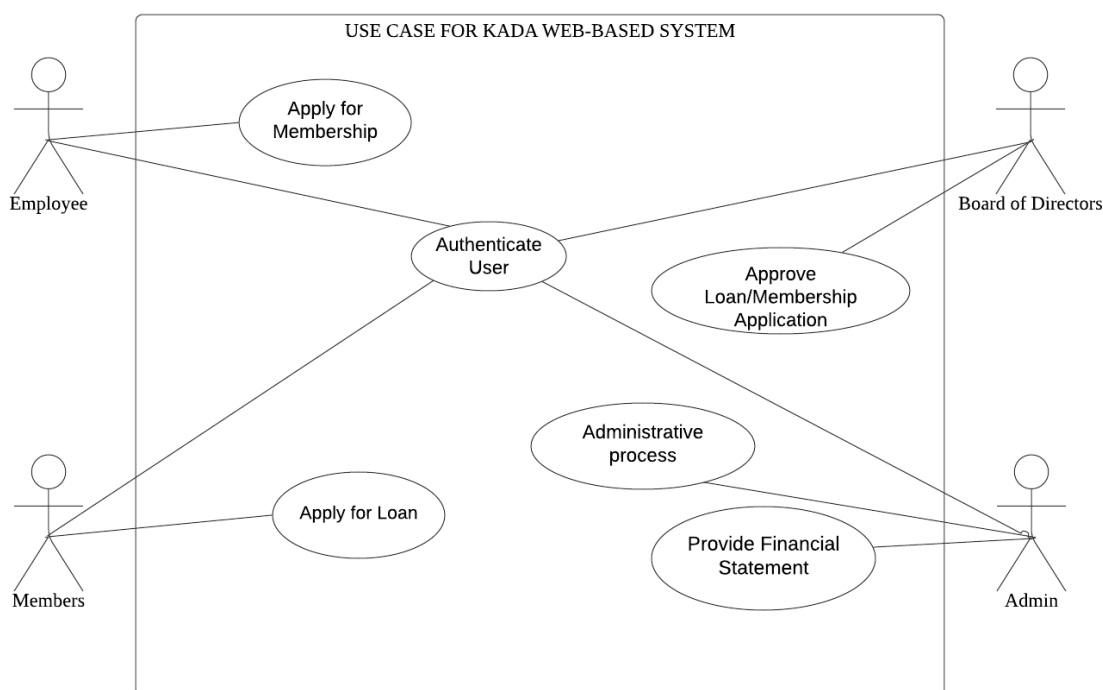
- As a member of the Board of Directors, I want to be able to log in to the system so that I can approve the loan applications.
- As a member of the Board of Directors, I want to be able to access the dashboard and look for any information I want.



## 1.7 System Features

The KADA's Cooperative web system is a software system that operates on both mobile devices and desktop ,including Android, iOS, Windows and Mac operating systems. The system provides a means for KADA employees to apply as a member of the cooperative to enjoy the benefits of the cooperative.

Figure 2.1 shows the use case diagram of the system that consists of the user of the system and their respective roles or tasks that can be done within the system.



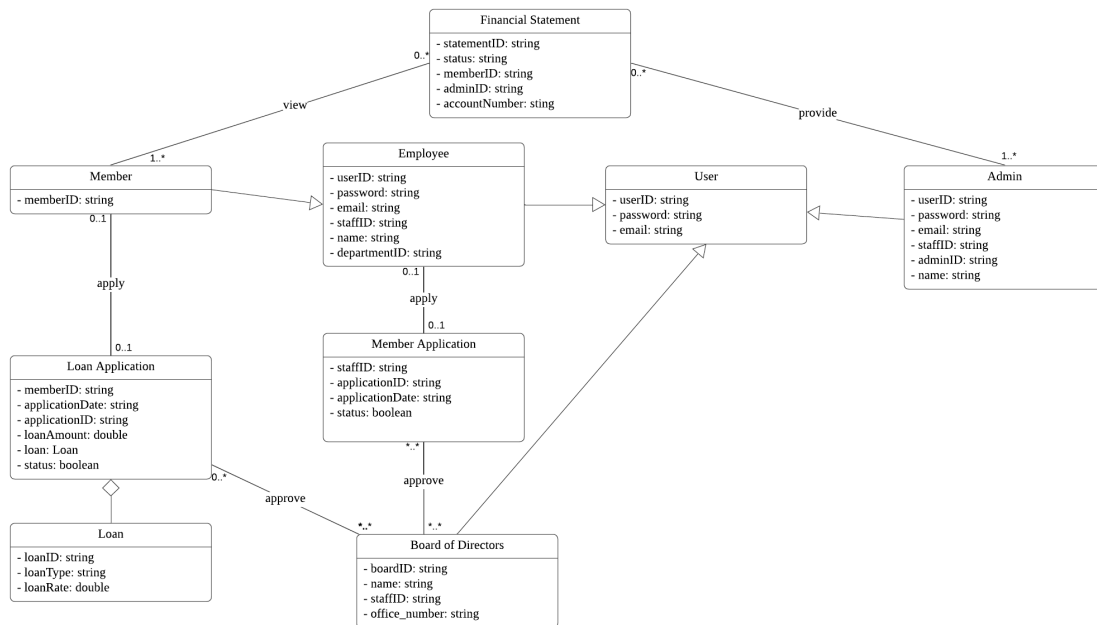
**Figure 2.1: Use Case Diagram for KADA's Cooperative Web-Based System**

Table 2.1 describes the task or use case from the use case diagram above.

**Table 2.1: Description of Module and Functions for KADA's Cooperative Web-Based System**

| Use case | Function                    | Description                                                                                                   |
|----------|-----------------------------|---------------------------------------------------------------------------------------------------------------|
| UC001    | Authenticate User           | This case allows employees, members, board of directors and administrators to be able to log into the system. |
| UC002    | Apply for loan              | This case allows members to be able to apply for a loan.                                                      |
| UC003    | Provide Financial Statement | This case allows administrators to be able to provide financial statements to both members and employees.     |
| UC004    | Administrative process      | This case allows administrators to be able to edit, delete and create the system's databases.                 |
| UC005    | Apply for Membership        | This case allows the board of directors to be able to apply for membership.                                   |
| UC006    | ?                           |                                                                                                               |

Figure 2.1 below shows the use class diagram or the domain model of the system that consists entities that exists in the system, their attributes, relationships and multiplicities among them.



**Figure 2.2: Domain Model for KADA's Cooperative Web-Based System**

## 1.8 Launch Phase

| <b>Sprint</b>                               | <b>Team members assigned</b> |
|---------------------------------------------|------------------------------|
| Sprint #1<br>User Authentication            | Muhammad Afiq Danial         |
| Sprint #2<br>Apply for membership           | Muhammad Alif Fathi          |
| Sprint #n<br>Apply for loan                 | Afif Shaqir Irfan            |
| Sprint #n<br>Administrative process         | Iman Abadi                   |
| Sprint 5<br>Approve Loan/Member Application | Afif Shaqir Irfan            |
| Sprint 6<br>Provide Financial Statement     | Muhammad Alif Fathi          |

## 1.9 User Story Details

Below is the list of all user stories based on the use case.

### 2.4.1 US001: Login

**Table 2.1: User Story Description for Employee Login**

| User story: Employee Login                                                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b> US001                                                                                                                                                                                 |
| <b>User Story Description</b><br>As an employee in KADA, I want to be able to log into the system so that I can use the website                                                                  |
| <b>Flow of events:</b><br>1. The employee open the official website<br>2. The employee enter their username and password on the login page<br>3. The system validate their username and password |
| <b>Alternative flow <i>n</i>:</b><br>The system detect an error and prompt to the employee to re-enter all the information                                                                       |
| <b>Precondition:</b><br>The employee already has the authenticated username and password                                                                                                         |
| <b>Postcondition:</b><br>The employee successfully sign into the website                                                                                                                         |
| <b>Exception flow:</b>                                                                                                                                                                           |

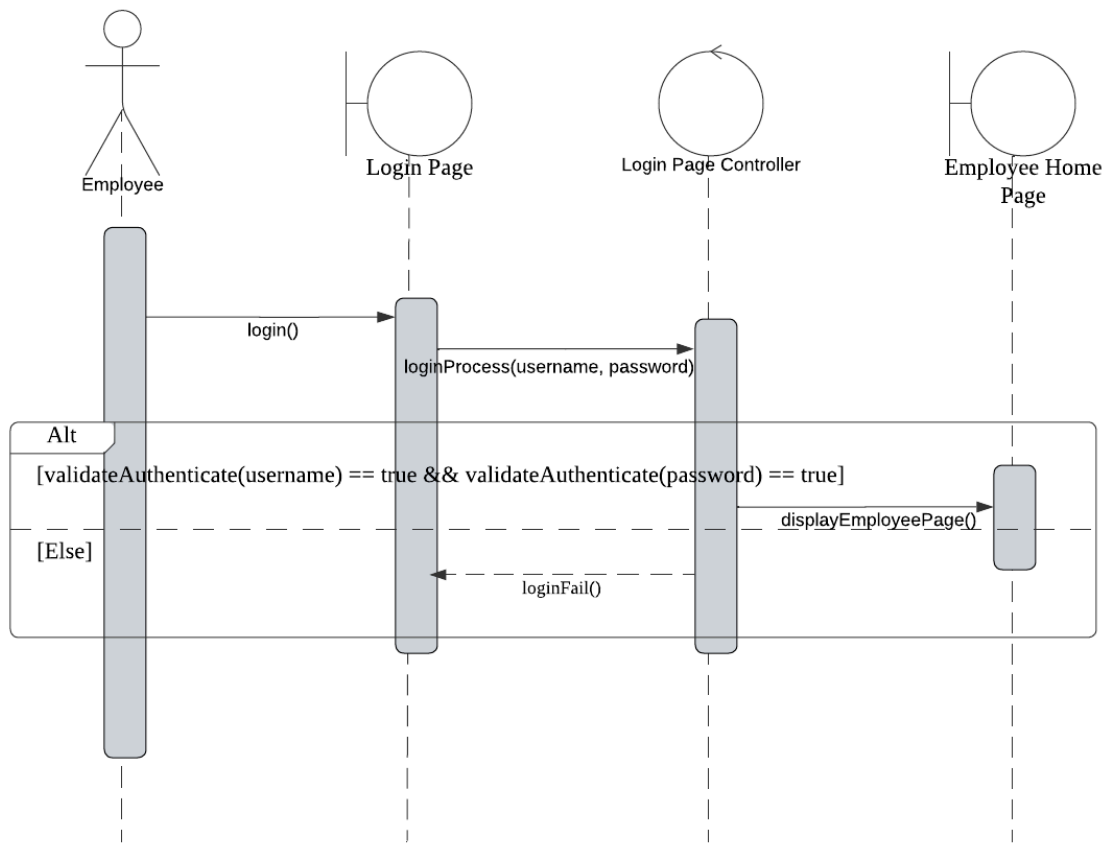


Figure x : Sequence Diagram for Employee Login Process

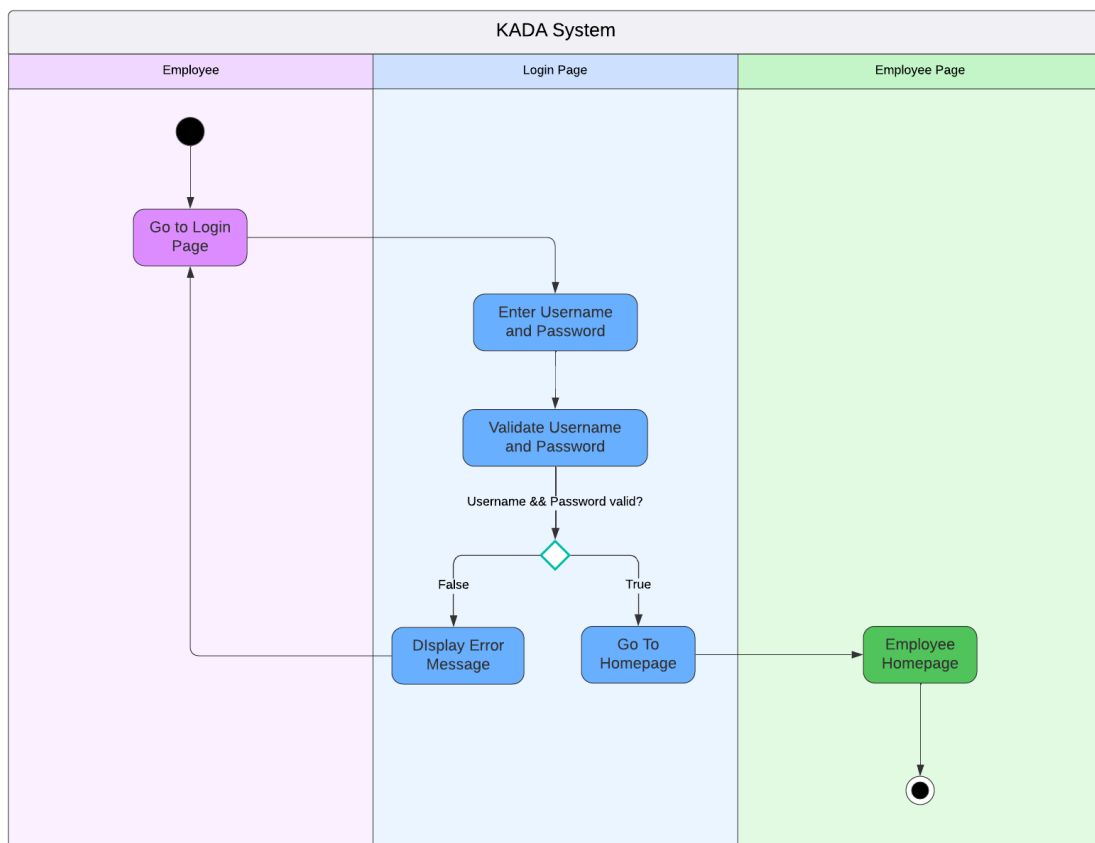


Figure x : Activity Diagram for Employee Login Process

**Table 2.2: User Story Description for Member Login**

| <b>User story: Member Login</b>                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b> US001                                                                                                                                                                                                                                    |
| <b>User Story Description</b><br>As a member of KADA, I want to be able to log into the system so that I can use the website.                                                                                                                       |
| <b>Flow of events:</b> <ol style="list-style-type: none"><li>1. The member open the official website</li><li>2. The member enter their username and password on the login page</li><li>3. The system validate their username and password</li></ol> |
| <b>Alternative flow 1:</b><br>The system detect an error and prompt to the member to re-enter all the information                                                                                                                                   |
| <b>Precondition:</b><br>The member already has the authenticated username and password.                                                                                                                                                             |
| <b>Postcondition:</b><br>The member successfully sign into the website                                                                                                                                                                              |
| <b>Exception flow:</b>                                                                                                                                                                                                                              |

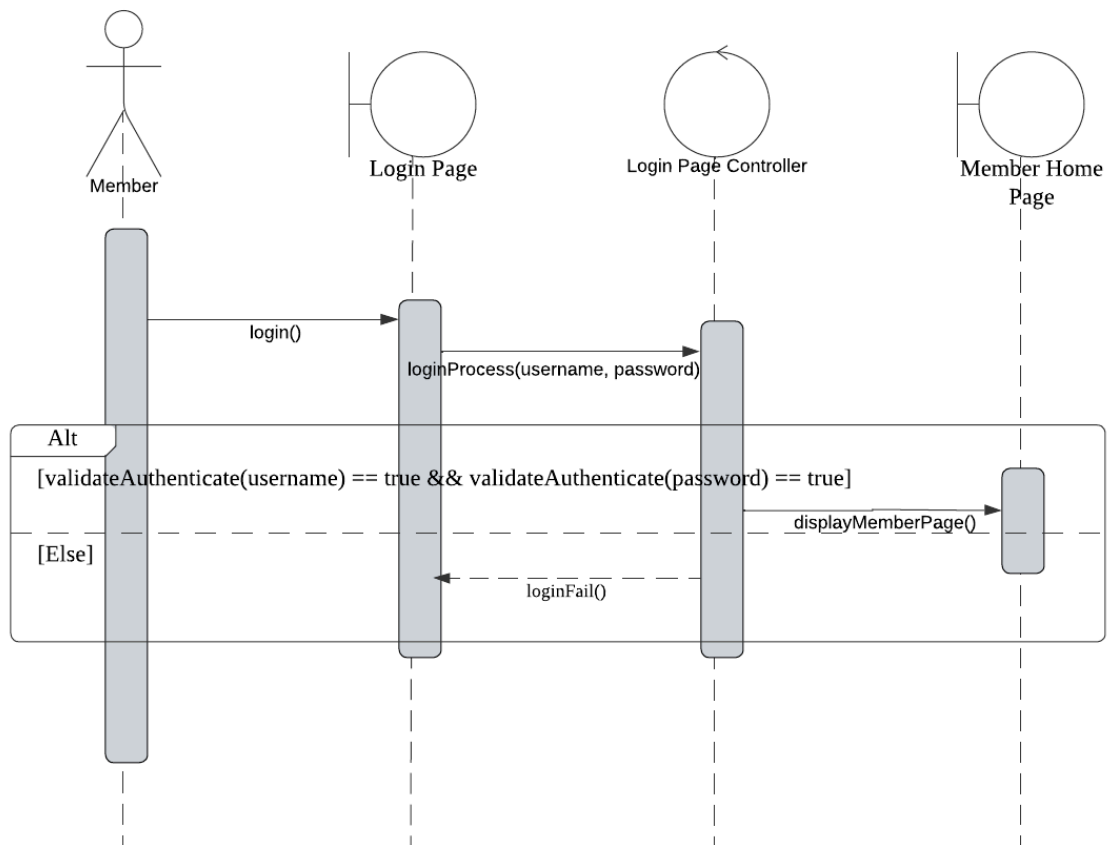


Figure x : Sequence Diagram for Member Login Process



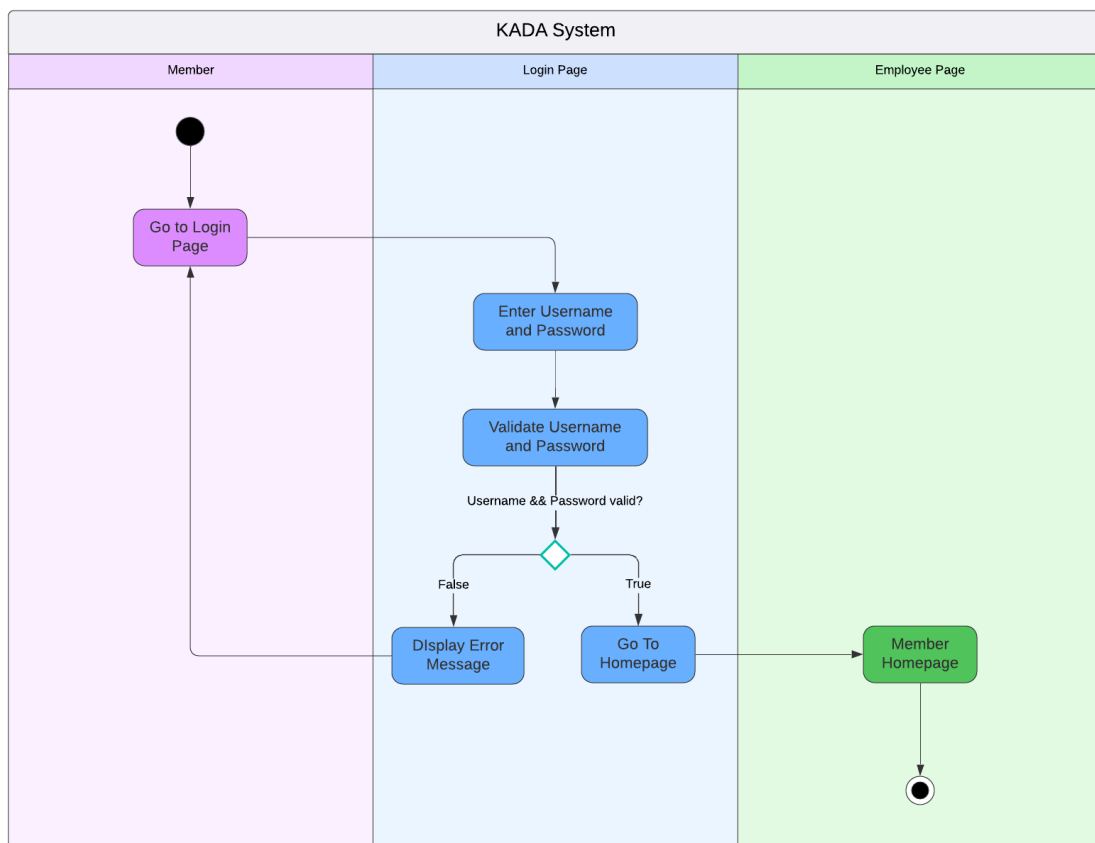
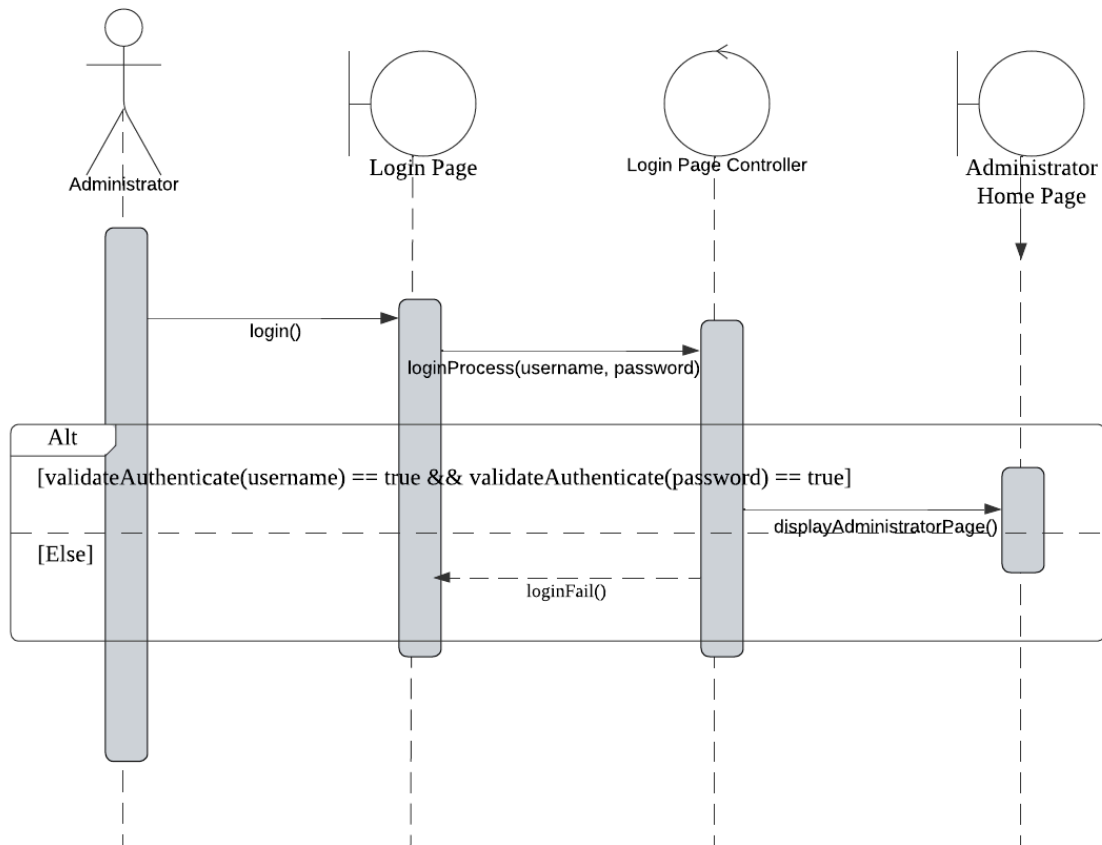


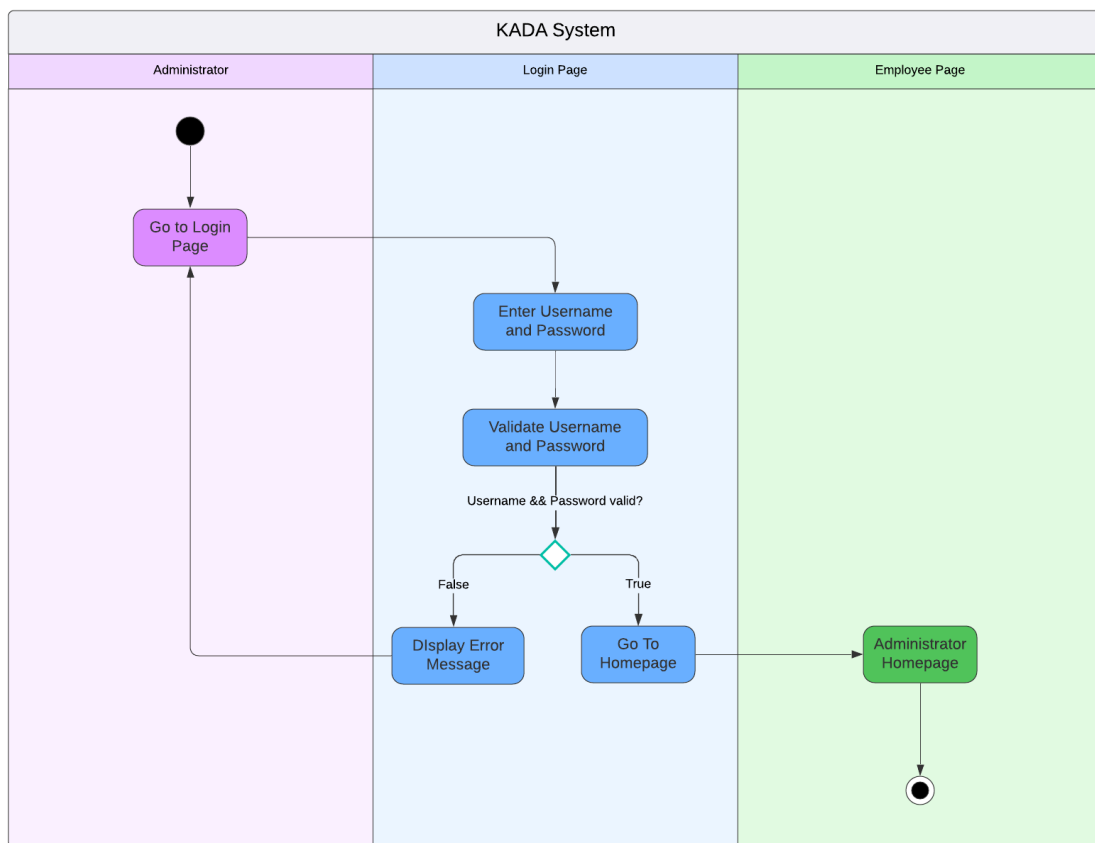
Figure x : Activity Diagram for Member Login Process

**Table 2.3: User Story Description for Administrator Login**

| User story: Administrator Login                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b> US001                                                                                                                                                                                                                                                  |
| <b>User Story Description</b><br>As an administrator of KADA, I want to be able to log into the system so that I can use the website.                                                                                                                             |
| <b>Flow of events:</b> <ol style="list-style-type: none"><li>1. The administrator open the official website</li><li>2. The administrator enter their username and password on the login page</li><li>3. The system validate their username and password</li></ol> |
| <b>Alternative flow 1:</b><br>The system detect an error and prompt to the administrator to re-enter all the information                                                                                                                                          |
| <b>Precondition:</b><br>The administrator already has the authenticated username and password.                                                                                                                                                                    |
| <b>Postcondition:</b><br>The administrator successfully sign into the website                                                                                                                                                                                     |
| <b>Exception flow:</b>                                                                                                                                                                                                                                            |



**Figure x : Sequence Diagram for Administrator Login Process**



**Figure x : Activity Diagram for Administrator Login Process**

**Table 2.3: User Story Description for Board of Directors Login**

| <b>User story: Board of Directors Login</b>                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b> US001                                                                                                                                                                                                                                                            |
| <b>User Story Description</b><br>As a Board of Directors of KADA, I want to be able to log into the system so that I can use the website.                                                                                                                                   |
| <b>Flow of events:</b> <ol style="list-style-type: none"><li>4. The Board of Directors open the official website</li><li>5. The Board of Directors enter their username and password on the login page</li><li>6. The system validate their username and password</li></ol> |
| <b>Alternative flow 1:</b><br>The system detect an error and prompt to the Board of Directors to re-enter all the information                                                                                                                                               |
| <b>Precondition:</b><br>The Board of Directors already has the authenticated username and password.                                                                                                                                                                         |
| <b>Postcondition:</b><br>The Board of Directors successfully sign into the website                                                                                                                                                                                          |
| <b>Exception flow:</b>                                                                                                                                                                                                                                                      |

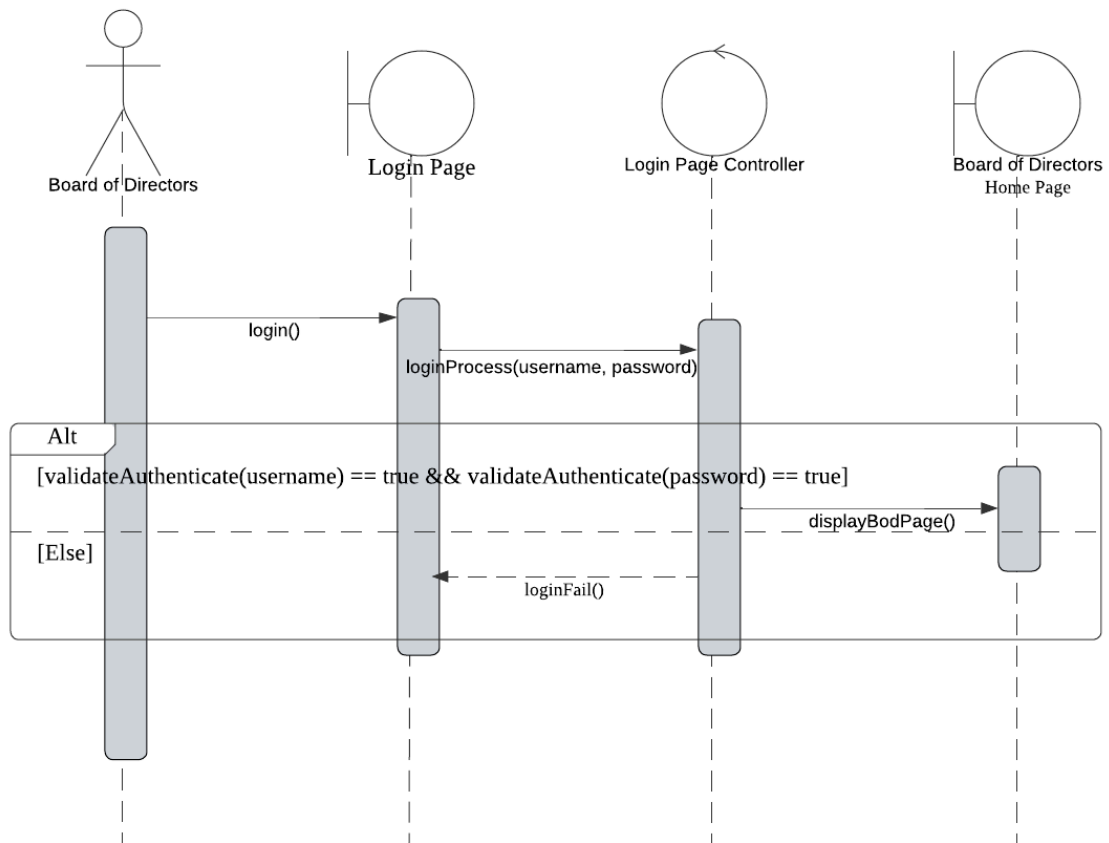
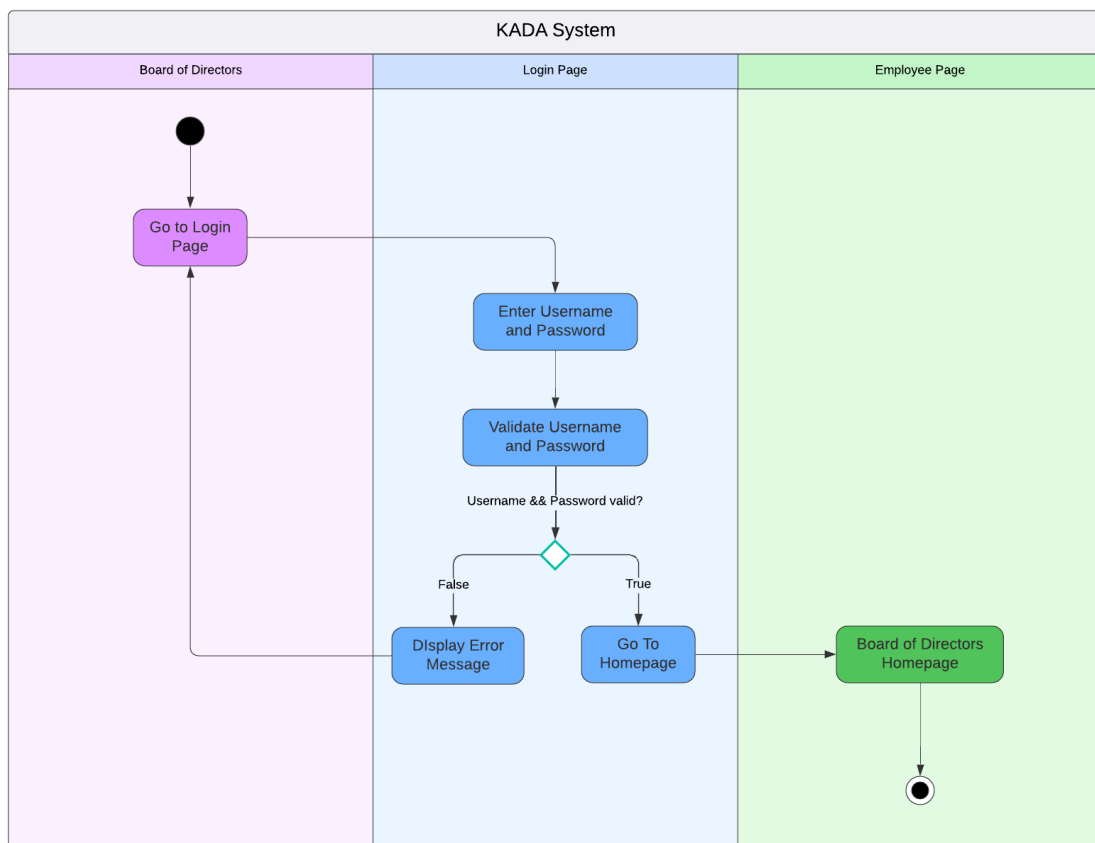


Figure x : Sequence Diagram for Board of Directors Login Process



**Figure x : Activity Diagram for Board of Directors Login Process**

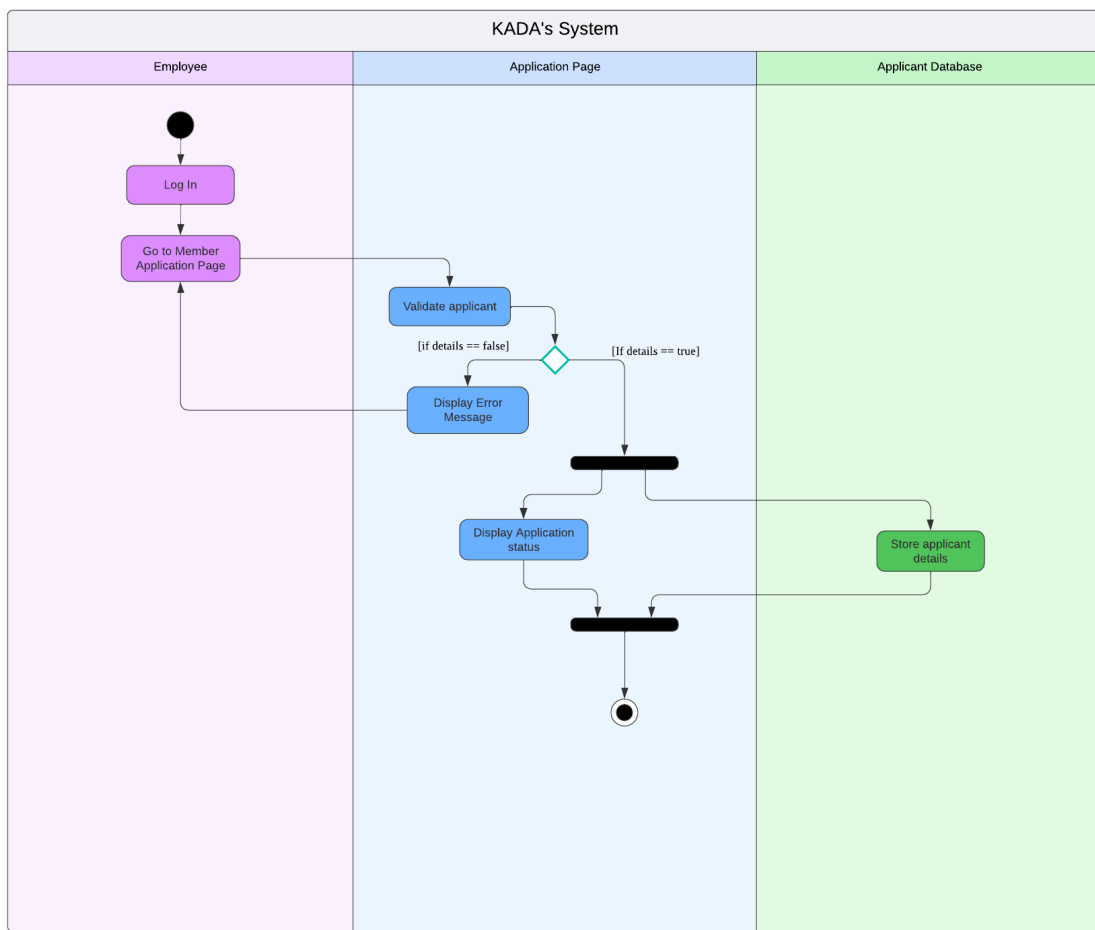
## 2.4.2 US002: Apply Membership

Table : User Story Description for Apply Membership

| User story: Apply Membership                                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ID: US002                                                                                                                                                                                                                                                                      |
| <b>User Story Description</b><br>As a employee of KADA, I want to be a member of KADA to have a permission to apply for personal loan                                                                                                                                          |
| <b>Flow of events:</b> <ol style="list-style-type: none"><li>1. The employee accesses the application page.</li><li>2. The employee key in all their personal details.</li><li>3. The system verified the input and waited for approval from the Board of Directors.</li></ol> |
| <b>Alternative flow :</b> <ol style="list-style-type: none"><li>1. If an employee fills in invalid details, the system will prompt an error message and members redirect to the application page.</li></ol>                                                                    |
| <b>Postcondition:</b><br>The employee successfully applied for a membership                                                                                                                                                                                                    |
| <b>Precondition:</b><br>The employee is already registered to the KADA membership.                                                                                                                                                                                             |
| <b>Exception flow:</b>                                                                                                                                                                                                                                                         |





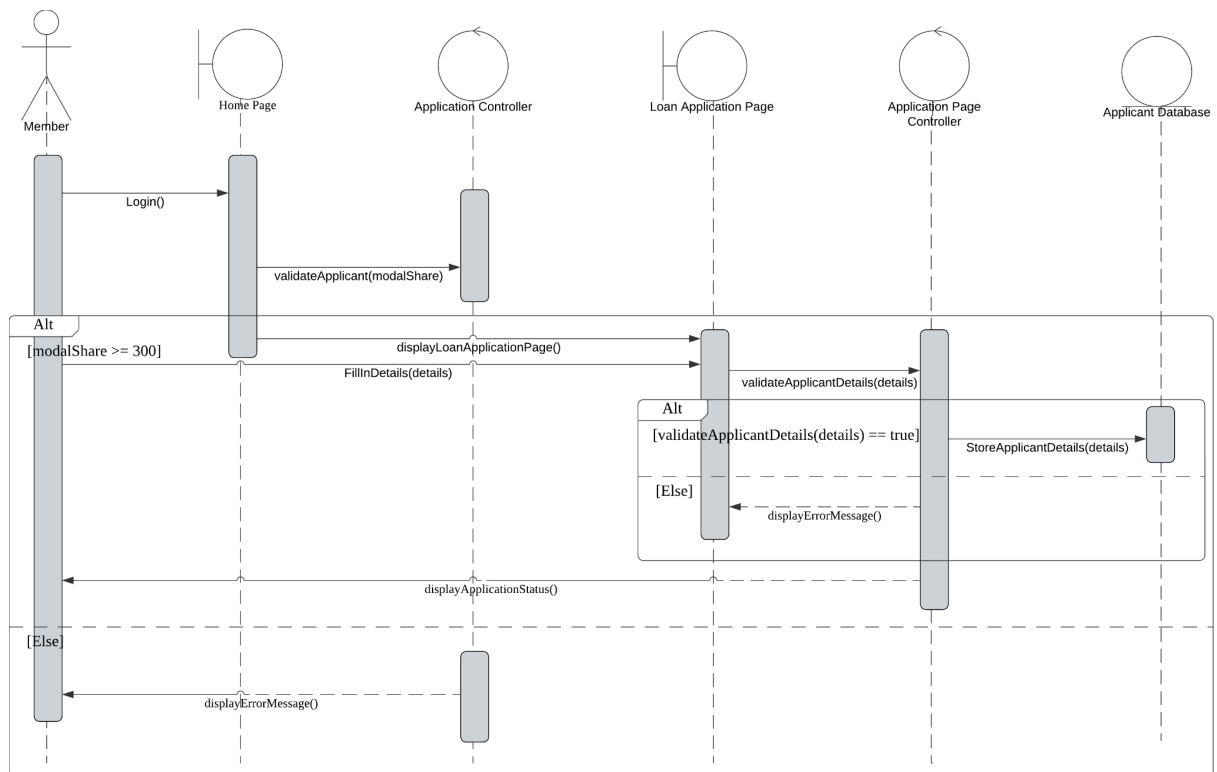


**Figure x : Activity Diagram for Applying Membership Process**

### 2.4.3 US003: Apply Loan

Table : User Story Description for Apply Loan

| User story: Apply Loan                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ID: US003                                                                                                                                                                                                                                                                                                                    |
| <b>User Story Description</b><br>As a member of KADA, I want to be able to access the loan application page so that I can apply for a loan.                                                                                                                                                                                  |
| <b>Flow of events:</b> <ol style="list-style-type: none"><li>4. The member accesses the application page.</li><li>5. The member key in their details (like amount of loan).</li><li>6. The system verified the input and waited for approval from the Board of Directors.</li></ol>                                          |
| <b>Alternative flow :</b> <ol style="list-style-type: none"><li>2. If members don't have modal shares more than or equal to 300, members cannot proceed with the application.</li><li>3. If members fill in invalid details, the system will prompt an error message and members redirect to the application page.</li></ol> |
| <b>Postcondition:</b><br>The member successfully applied for a loan.                                                                                                                                                                                                                                                         |
| <b>Precondition:</b><br>The member is already registered to the KADA membership.                                                                                                                                                                                                                                             |
| <b>Exception flow:</b>                                                                                                                                                                                                                                                                                                       |



**Figure x : Sequence Diagram for Member to Apply loan Process**

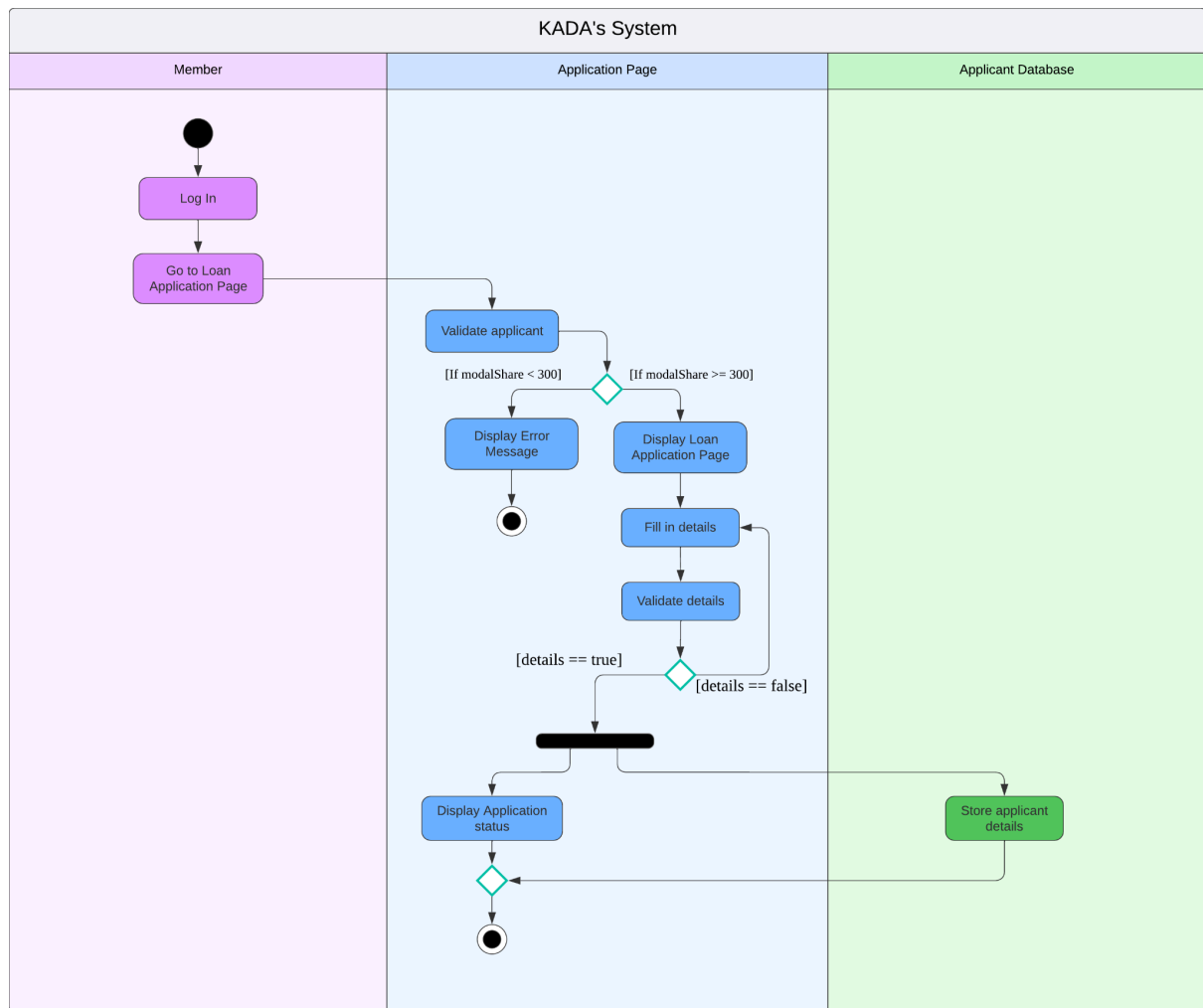
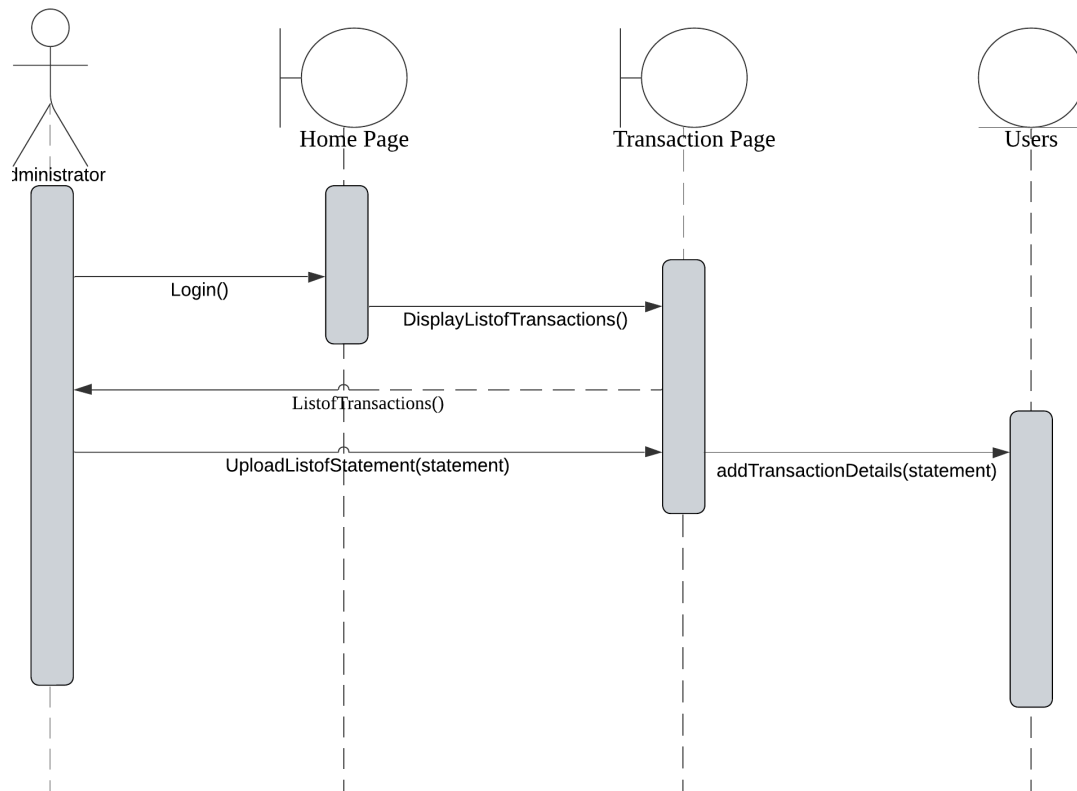


Figure x : Activity Diagram for Member to Apply Loan Process

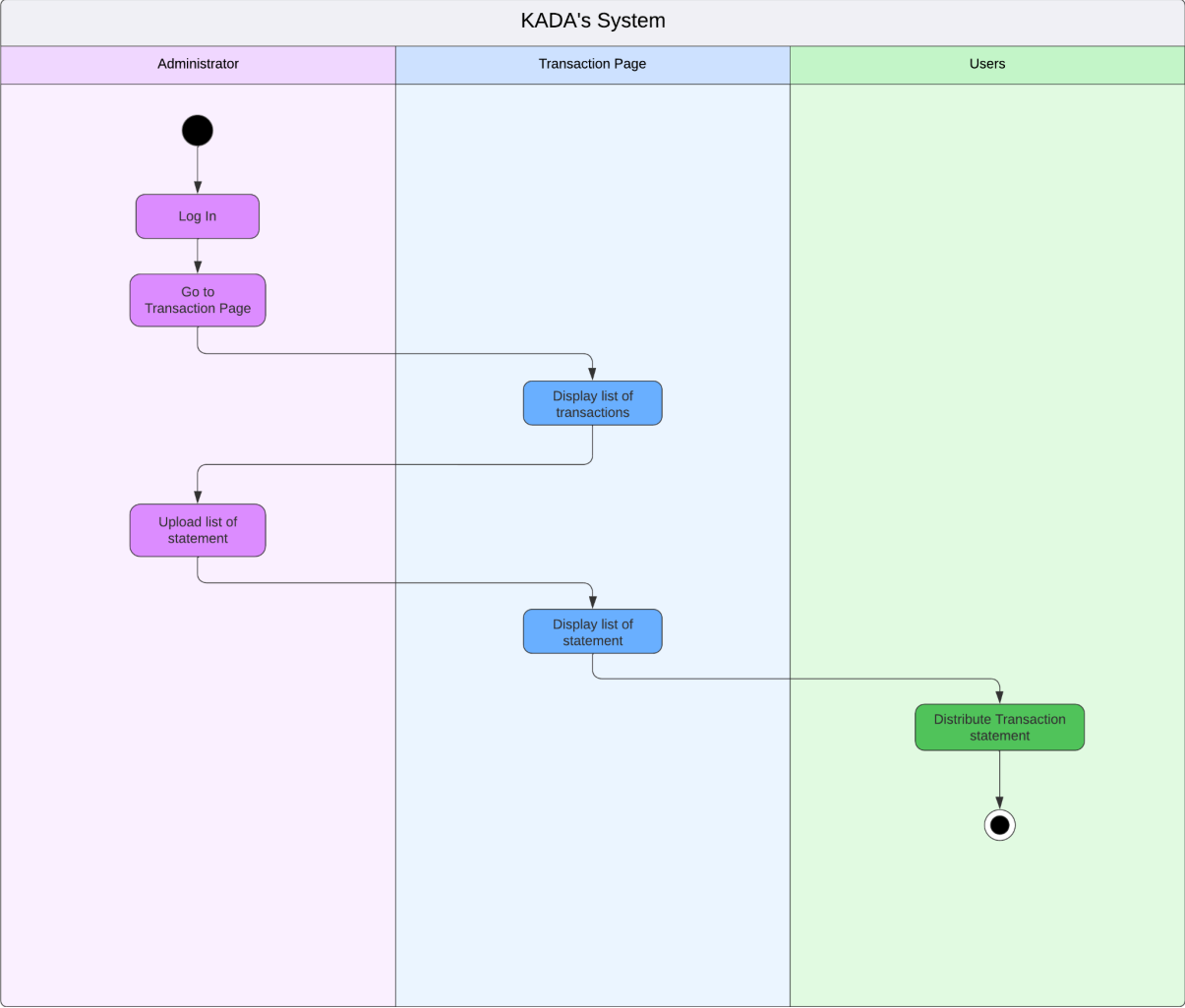
#### 2.4.4 US004: Provide Financial Statement

Table : User Story Description for Provide Financial Statement

| User story: Provide Financial Statement                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ID: US004                                                                                                                                                                                                            |
| <b>User Story Description</b><br>As a system administrator, I want to be able to access the transaction page so that I can use it to provide a financial statement and upload it into the database.                  |
| <b>Flow of events:</b> <ol style="list-style-type: none"><li>1. The admin accesses the transaction page.</li><li>2. The admin uploads the list of statements and adds the transaction details to the user.</li></ol> |
| <b>Alternative flow :</b>                                                                                                                                                                                            |
| <b>Postcondition:</b><br>The admin successfully provides a financial statement and uploads it into the database.                                                                                                     |
| <b>Precondition:</b><br>The admin is logged in as administrator.                                                                                                                                                     |
| <b>Exception flow:</b>                                                                                                                                                                                               |



**Figure x : Sequence Diagram for Administrator to Provide Financial Statement Process**



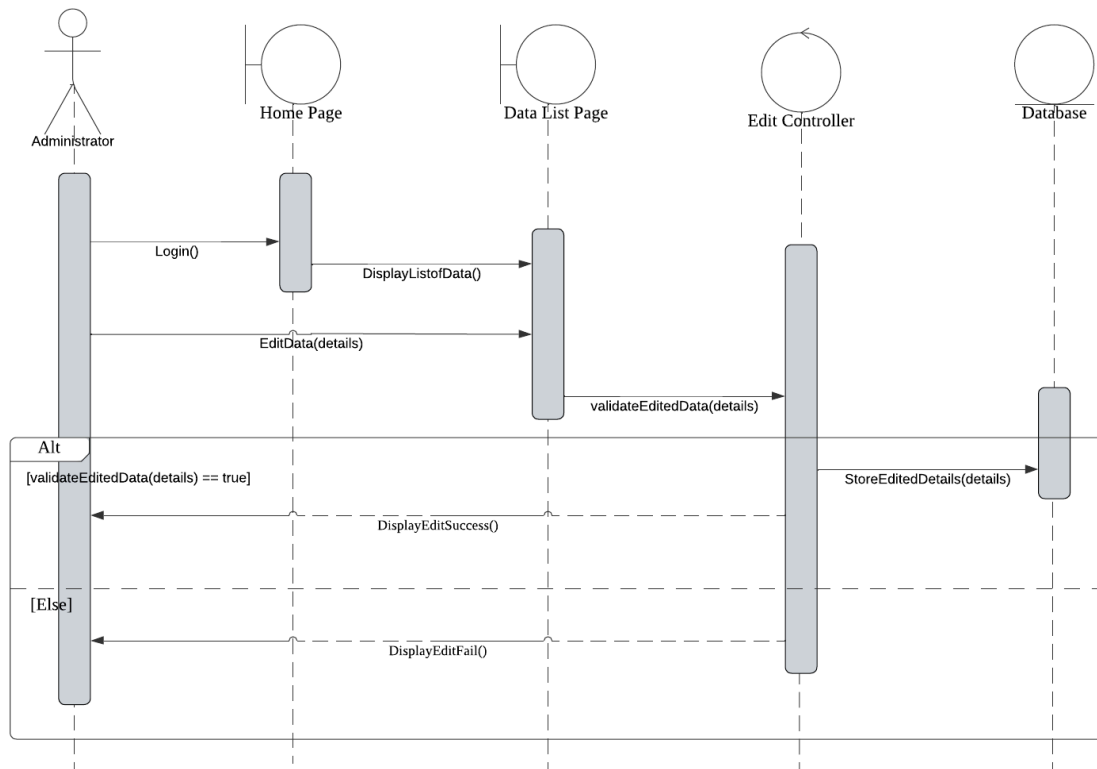
**Figure x : Activity Diagram for Administrator to Provide Financial Statement Process**



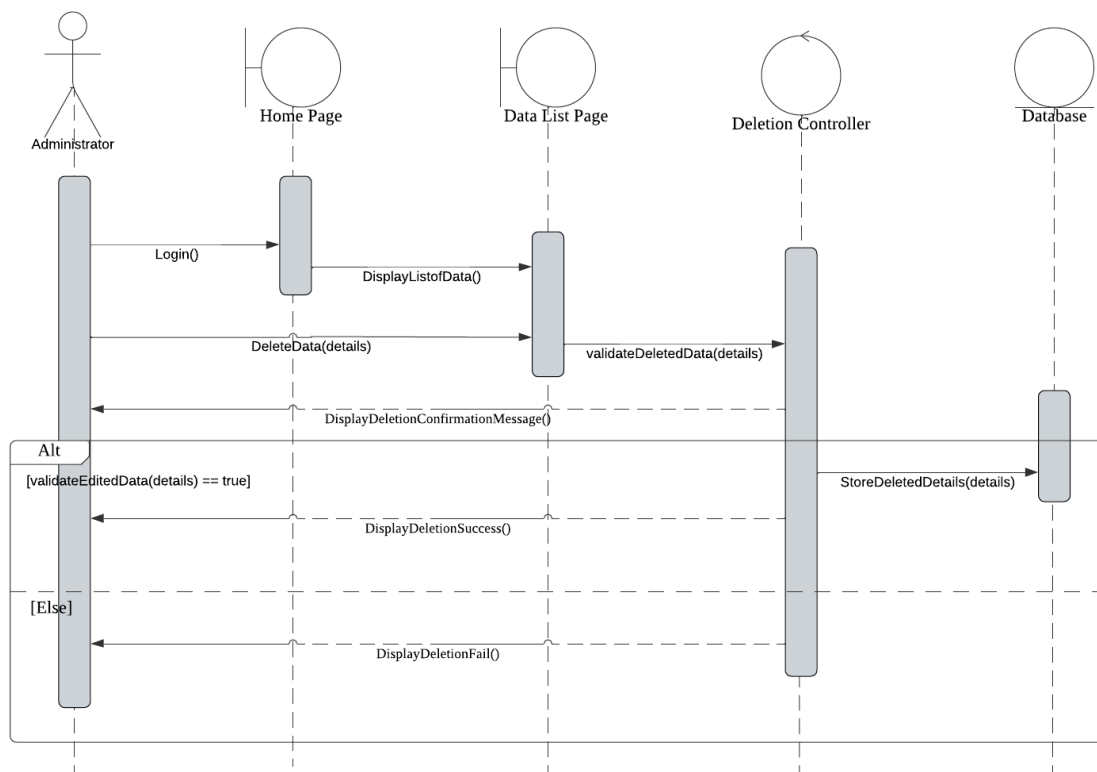
## 2.4.5 US005: Administrative Process

Table : User Story Description for Administrative Process

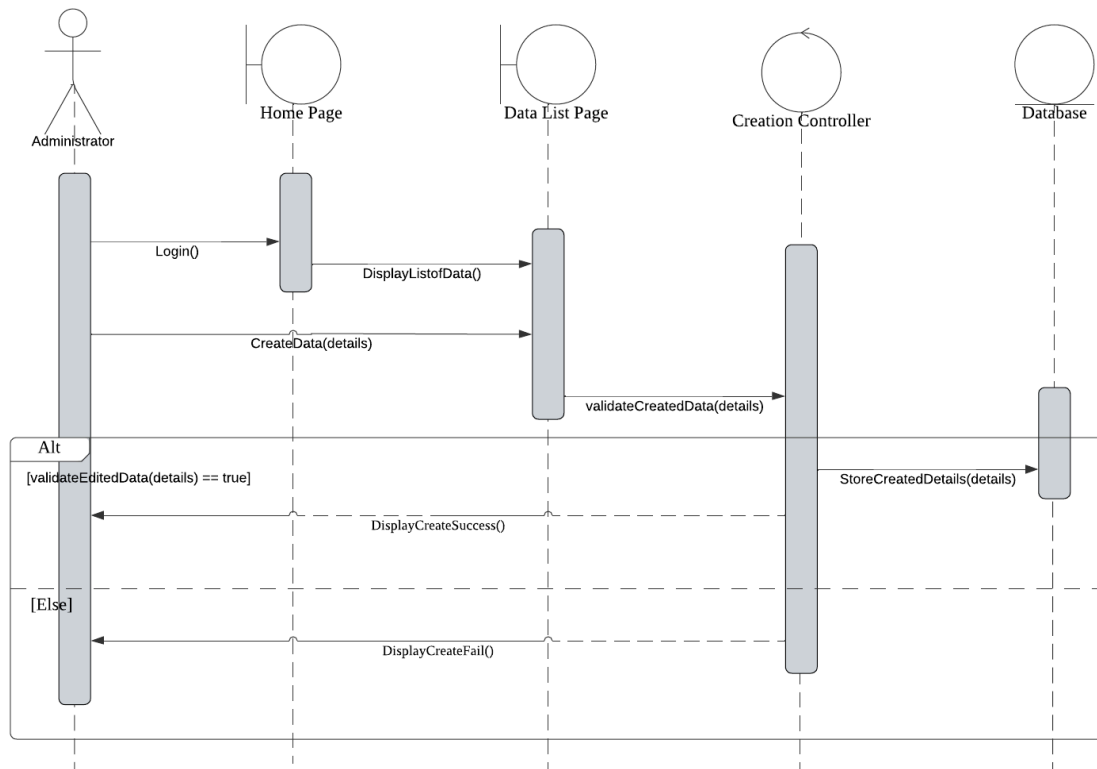
| User story: Administrative                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID:</b> US005                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>User Story Description</b><br><br>As a system's administrator, I want to be able to log into the system so that I can manage the data in the system's database.                                                                                                                                                                                                                                                                       |
| <b>Flow of events:</b> <ol style="list-style-type: none"><li>1. The admin views tables present in the database and the system will show the attributes and values according to the tables.</li><li>2. The admins click on the record that they want to manipulate.</li><li>3. The admin can create, update or delete the data that is in the database.</li><li>4. The system will display a message of successful data update.</li></ol> |
| <b>Alternative flow :</b><br>Duplicated data: The system prompts of invalid data and asks to re-enter information                                                                                                                                                                                                                                                                                                                        |
| <b>Postcondition:</b><br>The database is updated successfully.                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Precondition:</b><br>The user is logged in as an administrator.                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Exception flow:</b>                                                                                                                                                                                                                                                                                                                                                                                                                   |



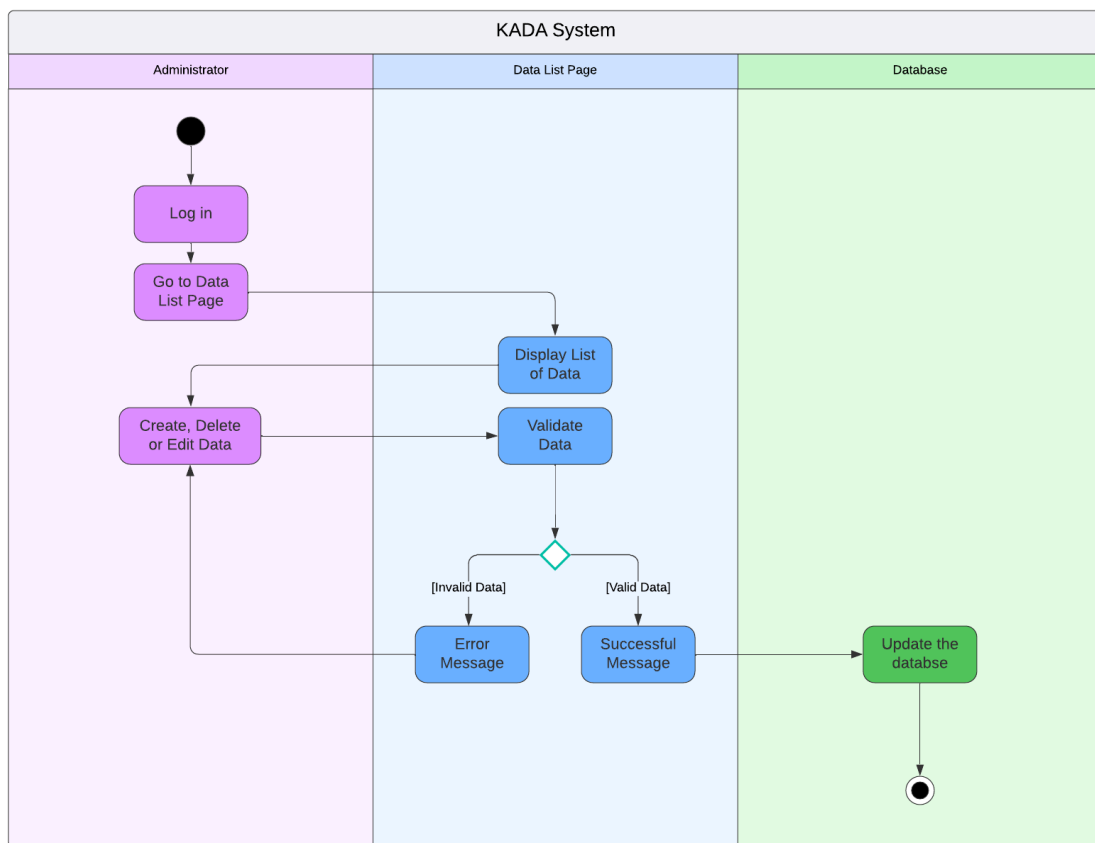
**Figure x : Sequence Diagram for Administrative Process for Editing Data/Record**



**Figure x : Sequence Diagram for Administrative Process for Deleting Data/Record**



**Figure x : Sequence Diagram for Administrative Process for Creating New Data/Record**



**Figure x : Activity Diagram for Administrator doing Administrative Process**

## 2.4 Performance and Other Requirements

Software system attributes define the overall qualities or characteristics of the software. These attributes are the foundation on which the system is built, and they include the following:

- **Correctness:** The system shall be able to correctly behave as per the system requirements, specifications and features.
- **Maintainability:** The system shall be able to be maintained easily by adding new features and updating existing features with new technologies.
- **Usability:** The system shall be able to be usable towards the users of the system for a good customer experience.
- **Portability:** The system shall be portable where the system can be migrated to other environments with different hardware and operating systems.
- **Compatibility:** The system shall be able to be compatible with multiple operating systems and web browsers without affecting intended functionality.

## 2.5 Design Constraints

Below are the constraints imposed by the organisation.

- Environmental constraints: The system must be designed to operate in cold temperatures for the database to work fine. The temperature must be 0-25 Celcius.
- Hardware constraints: The system must be designed to run with enough storage and RAM because the system will run on KADA's on-premises or cloud-based server setup. The processors used are Intel Xeon or AMD EPYC with a minimum of 8 cores. The minimum ram is 16 GB or more for optimal performance. Solid State Drive (SSD) with at least 512 GB for fast data retrieval and secure storage and minimum bandwidth of 1 Gbps.
- Security constraints: The system must be designed to encrypt all the user data with secure protocol using AES-256 for storage and SSL/TLS for transmission. For the communication, the system used HTTPS for secure web-based access to the cooperative system and SMTP for sending and receiving emails within the KADA users.
- Compatibility constraints: The system must be designed to be compatible with many operating systems like Windows and IOS. This will help KADA employees to work with ease.
- Performance constraints: The system must be designed to support a powerful networking environment with support a maximum of 1200 concurrent users with low response time which is less than 2 seconds.